(i=2) → visited = False

3 > 2 → swap → [2, 3, 5, 6, 7] → visited = true

3 > 5 → no swap

end: [2, 3, 5, 6, 7]

(i=3) → visited = False

2 > 3 → no swap

end: [2, 3, 5, 6, 7] → visited = False → Break

∴ [2, 3, 5, 6, 7]

---

(1) [13, 46, 24, 53, 21, 19]

→ [13, 46, 24, 53, 21, 19
→ [13, 24, 46, 53, 21, 19]
→ [13, 24, 46, 53, 21, 19]
→ [13, 24, 46, 21, 53, 19]
→ [13, 24, 46, 21, 19, 53]

n-1

(2) [13, 24, 46, 21, 19, 53]

→ [13, 24, 46, 21, 19, 53]
→ [13, 24, 46, 21, 19, 53]
→ [13, 24, 21, 46, 19, 53]
→ [13, 24, 21, 19, 46, 53]

n-2

```
int [] arr = {13, 46, 24, 53, 21, 19};
int n = arr.length;

for (int i=0; i<n-1; i++){
    boolean visited = false;
    for (int
```

③ [13, 24, 21, 19, 46, 53]

→ [13, 24, 21, 19, 46, 53]
→ [13, 21, 24, 19, 46, 53]
→ [13, 21, 19, 24, 46, 53]

n - 3

④ [13, 21, 19, 24, 46, 53]

→ [13, 21, 19, 24, 46, 53]
→ [13, 19, 21, 24, 46, 53]

n - 4

⑤ [13, 19, 21, 24, 46, 53]

→ [13, 19, 21, 24, 46, 53]

n - 5

```
for (int i=0; i<n-1; i++){
    boolean visited = false;
    for (int j=0; j<n-i-1; j++){
        if(arr[j] > arr[j+1]){
            int temp = arr[j];
            arr[j] = arr[j+1];
            arr[j+1] = temp;
            visited = true;
        }
    }
    if(!visited){
        break;
    }
```

arr = {3, 7, 5, 2, 6};
n = 5

Outer loop : for (int i=0; i<n-1; i++)
0 → 3

For : $\boxed{i=0}$
boolea visited = false

Inner loop : for (int j=0; j<n-i-1; j++)
$\boxed{j=0} \rightarrow \boxed{j=3}$

For : $\boxed{j=0}$
3 > 7 → no swap

For : $\boxed{j=1}$
7 > 5 → swap → visited = True
[3, 5, 7, 2, 6]

For $\boxed{j=2}$
7 > 2 → swap
[3, 5, 2, 7, 6]

For $\boxed{j=3}$
7 > 6 → swap
[3, 5, 2, 6, 7]

∴ [3, 5, 2, 6, 7]

[3, 5, 2, 6, 7]

For i = 1 → be visited = False

Inner Loop: for (int j = 0; j < n-i-1; j++)
j = 0 → j = 2

For j = 0:
3 > 5 → no Swap

For j = 1:
5 > 2 → Swap → visited = true
[3, 2, 5, 6, 7]

For j = 2:
5 > 6 → no swap

∴ [3, 2, 5, 6, 7] → visited = True → continue;

For i = 2 → visited = False [3, 2, 5, 6, 7]

Inner Loop: for (int j = 0; j < n-i-1; j++)
1 < 2 5-2-1
j = 0 → j = 1

For j = 0:
3 > 2 → Swap → visited = True
[2, 3, 5, 6, 7]

For j = 1:
3 > 5 → no Swap

∴ [2, 3, 5, 6, 7] → visited = True → Continue.

[2, 3, 5, 6, 7]

For i = 3 → visited = False 4

5 - 3 - 1

Inner Loop : for (int j = 0; j < n - i - 1; j++)

j = 0 :

For j = 0 :

2 > 3 → no Swap

visited = False break

∴ Final ans = [2, 3, 5, 6, 7]